

# MultiFEBE

## Examples

This document lists all the included examples for users, and it also describes the guidelines for developing and including new examples.

Examples are organized by the type of problem **PP**, type of analysis **AA**, modeling **MM**, and number **NNN**, i.e. a generic example is encoded as **PP-AA-MM-NNN**. The examples whose title starts with “[TUTORIAL]” include a document with a description of the modeling approach, input files, output files and a comparison of the obtained results against other reference results (analytical or otherwise), which serves as validation.

## 1 List of examples for users

- **ME** Linear elastic mechanics
  - **ST** Static
    - \* **SR** Structures
      - 001 [TUTORIAL] Cantilever beam (FEM model).
      - 002 [TUTORIAL] Rectangular Plate (FEM model).
      - 003 [TUTORIAL] Scordelis-Lo Roof (FEM model).
      - 004 [TUTORIAL] Spherical Shell (FEM model).
    - EL** Elastostatics (continuum)
      - 001 [TUTORIAL] Plain strain square (BEM model).
      - 002 [TUTORIAL] Cube (BEM model).
    - \* **CO** Coupled
      - 001 [TUTORIAL] Floating pile response under head force or moment (BEM-FEM model).
  - **TH** Time Harmonic
    - \* **SR** Structures
      - 001 [TUTORIAL] Cantilever beam (FEM model).
    - \* **EL** Elastodynamics (continuum)
      - 001 [TUTORIAL] Cube (BEM model).
      - 002 [TUTORIAL] Cantilever wall (BEM model).
      - 003 [TUTORIAL] Soria arch dam: fixed-base (BEM model).
    - \* **PO** Poroelastodynamics (continuum)
      - **none**
    - \* **AC** Acoustics (continuum)
      - 001 [TUTORIAL] Pressure waves in a room (BEM model).
    - \* **CO** Coupled
      - 001 [TUTORIAL] Impedances of inclined pile foundations (BEM-FEM model).
      - 002 [TUTORIAL] Seismic response of a single inclined pile (BEM-FEM model).

- 003 [TUTORIAL] Impedances of a suction caisson / bucket foundation (BEM-FEM model).
- 004 [TUTORIAL] Seismic response of an Offshore Wind Turbine (BEM-FEM model).
- 005 [TUTORIAL] Soria arch dam: compliant base (BEM model).
- 006 [TUTORIAL] Soria arch dam: compliant base with 112 m water depth (BEM model).

## 2 Notes for developers

Examples should be categorized within the following tree:

- ME Linear elastic mechanics
  - ST Static
    - \* SR Structures. Examples using only structural and discrete finite elements.
    - \* EL Elastostatics (continuum). Examples using either boundary elements or finite element, where the domain is discretized without structural simplifications, i.e. a continuum or solid model.
    - \* CO Coupled. Examples where structures are coupled with the surrounding media.
  - TH Time Harmonic
    - \* SR Structures. Examples using only structural and discrete finite elements.
    - \* EL Elastodynamics (continuum). Examples using either boundary elements or finite element, where the domain is discretized without structural simplifications, i.e. a continuum or solid model. Only elastic solids.
    - \* PO Poroelastodynamics (continuum). The same as above, but when using poroelastic media.
    - \* AC Acoustics (continuum). The same as above, but when using inviscid fluids.
    - \* CO Coupled. Examples where structures are coupled with the surrounding media, and/or where different material models are interacting.
- LA Laplace (to be developed)

Within the **examples** folder, the example must have a dedicated folder with the name **PP-AA-MM-NNN** according to its categorization and numbering. The structure within the example folder must be:

- **PP-AA-MM-NNN.pdf** - It is a mandatory document describing example. It can be as simple as a readme type of document, where a brief description of the example is given, or it can be a fully documented example in a tutorial format.
- **doc\_src** - It is a mandatory folder containing the sources for creating **PP-AA-MM-NNN.pdf**. It can be a plain text file **PP-AA-MM-NNN.txt**, a Microsoft Word file **PP-AA-MM-NNN.doc** or **PP-AA-MM-NNN.docx**, a Open Document Text file **PP-AA-MM-NNN.odt**, or a LaTeX file **PP-AA-MM-NNN.tex**. The preferred document source is LaTeX.
- **case\_files** - It is a mandatory folder containing all input files, including MultiFEBE, mesh and other files required for running the example.